

Homework #0 - Aaron's Solutions

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A1

$$TP = 0.99 \cdot \frac{1}{10,000}$$

$$FP = 0.01 \cdot \frac{9,999}{10,000}$$

Therefore, the probability that you actually have the disease given a positive test result is

$$\frac{TP}{TP + FP} = \frac{0.99 \cdot \frac{1}{10,000}}{0.99 \cdot \frac{1}{10,000} + 0.01 \cdot \frac{9,999}{10,000}} = \frac{1}{102}.$$